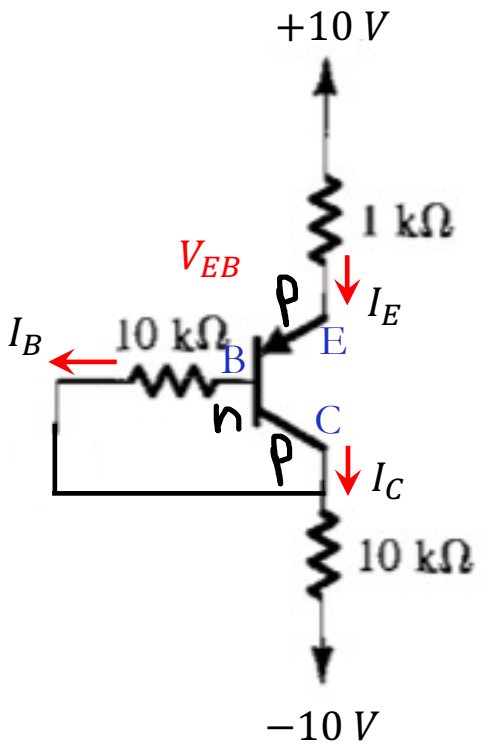
Bipolar Junction Transistors

- DC Analysis (Biasing) of BJTs.
- IV and Transfer Function curves.

Examples

5) Analyze the DC BJT circuit below. (β=50).



Assume active mode

$$V_{EB} = 0.7\ V$$

KVL in the (+10) to (-10) branch

$$10 - I_E(1\ k) - V_{EB} - \frac{I_E}{\beta + 1}(10\ k) - I_E(10\ k) + 10 = 0$$

$$I_E = 1.38\ mA$$

$$V_E = 10 - (1.38\ m)(1\ k) = 8.62\ V$$

$$V_B = V_E - V_{EB} = 7.92\ V$$

$$V_C = (1.38\ m)(10\ k) - 10 = 3.8\ V$$

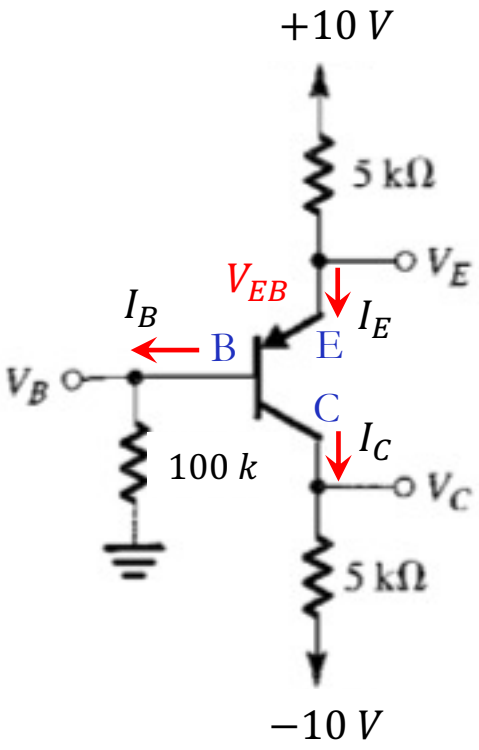
$$V_C < V_B$$

CB junction is reversed biased

Assumption is correct

Examples

6) Find the parameters α and β of the transistor below given one measurement.



$$V_B = +1\ V$$

$$V_E = +1.7\ V$$

$$I_E = \frac{10 - 1.7}{5\ k} = 1.66\ mA$$

$$I_B = \frac{1}{100\ k} = 10\ \mu A$$

$$\beta = \frac{1.66\ mA}{10\ \mu} - 1 = 165$$

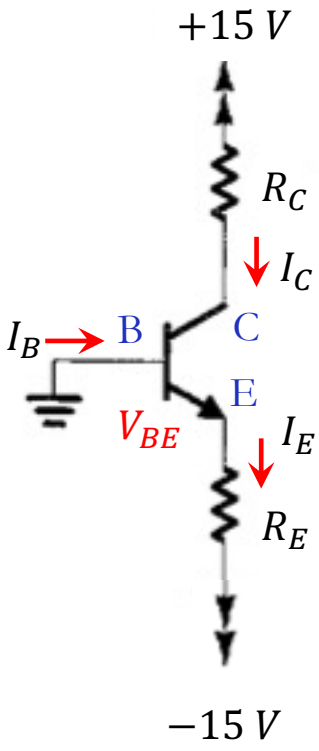
$$\alpha = \frac{165}{165 + 1} = 0.994$$

$$V_C = 0.994(1.66\ m)(5\ k) - 10 = -1.75\ V < V_B$$

Active mode

Examples

7) Design the circuit for I_C=2 mA and V_C=5 V. (β=100).



As given, the circuit is in active mode.

$$V_C = 5\ V$$

$$V_B = 0\ V$$

$$V_E = 0 - 0.7 = -0.7\ V$$

$$I_C = 2\ mA \rightarrow R_C = \frac{15-5}{2\ m} = 5\ k\Omega$$

$$I_E = \alpha\ I_C = \frac{100}{101}\ 2\ m = 2.02\ mA$$

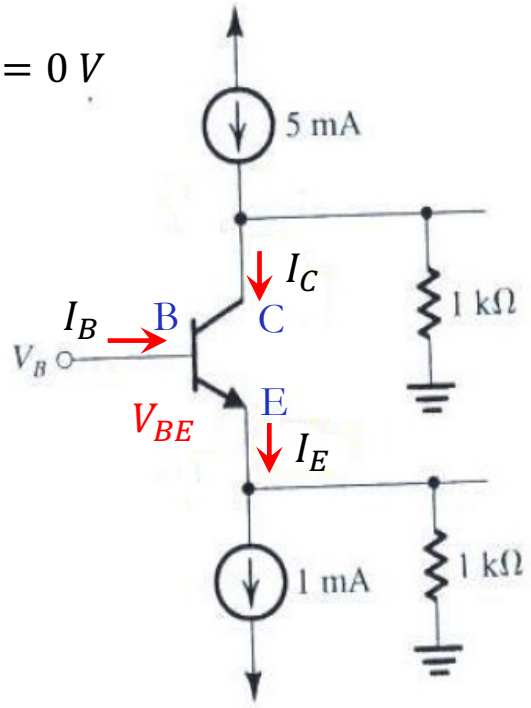
$$R_E = \frac{-0.7-(-15)}{2.02\ m} = 7.07\ k\Omega$$

Examples

8) Analyze the DC BJT circuit below. (β=100).

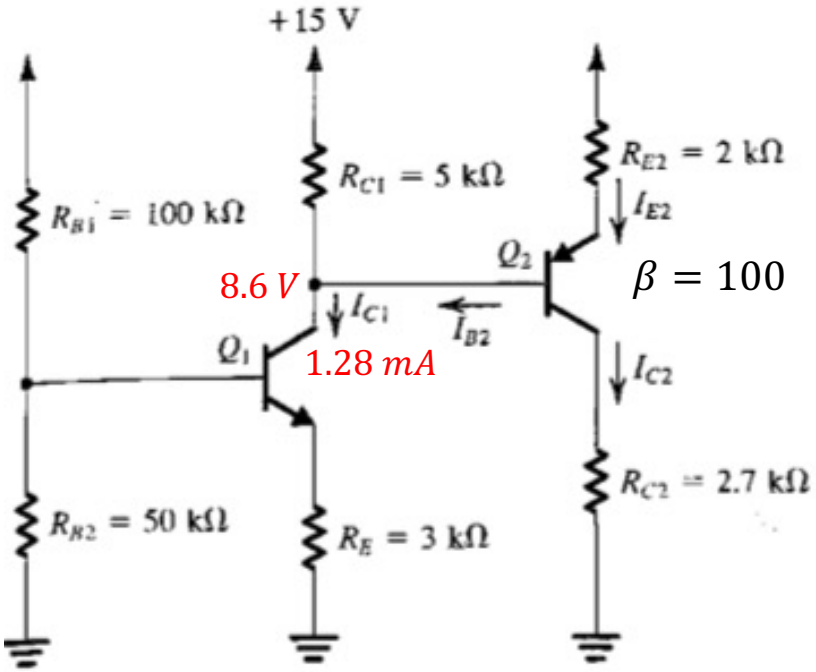
$V_B = 1\ V$

$V_B = 0\ V$



Examples

9) Another transistor stage is added to example (4).



$\beta = 100$

$$V_{B2} = V_{C1} = 8.6\ V$$

$$V_{E2} = 8.6 + 0.7 = 9.3\ V$$

$$I_{E2} = \frac{15 - 9.3}{2\ k} = 2.85\ mA$$

$$I_{C2} = 2.82\ mA$$

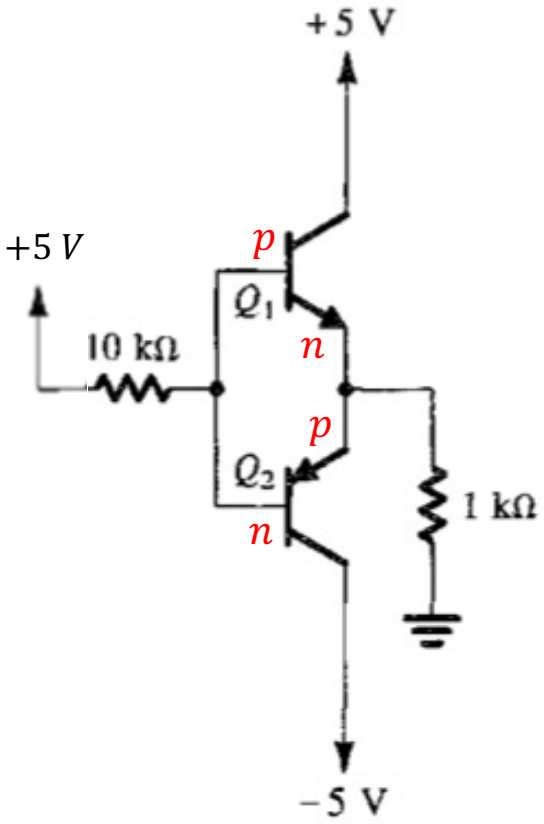
$$V_{C2} = (2.82\ m)(2.7\ k) = 7.62\ V$$

Active mode

$$I_{B2} = 28.2\ \mu A$$

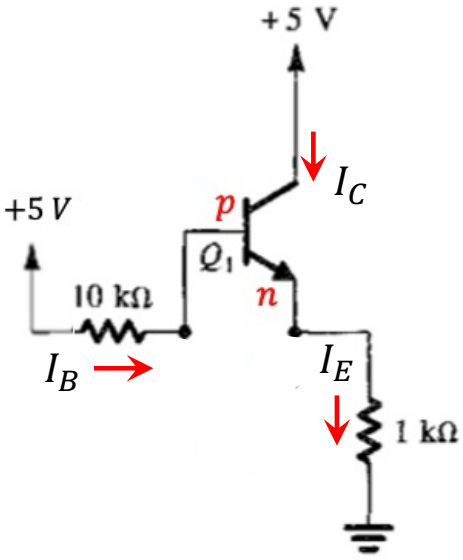
Examples

10) Analyze the DC BJT circuit below. (β=100).



The two transistors cannot be both ON, because the EB junctions are connected across each other.

With the given biasing, Q_2 is OFF.



Assume active mode

$$V_{BE} = 0.7\ V$$

KVL in the B-E circuit

$$5 - I_B(10\ k) - 0.7 - (\beta + 1)I_B(1\ k) = 0$$

$$I_B = 38.7\ \mu A$$

$$I_C = 3.87\ mA$$

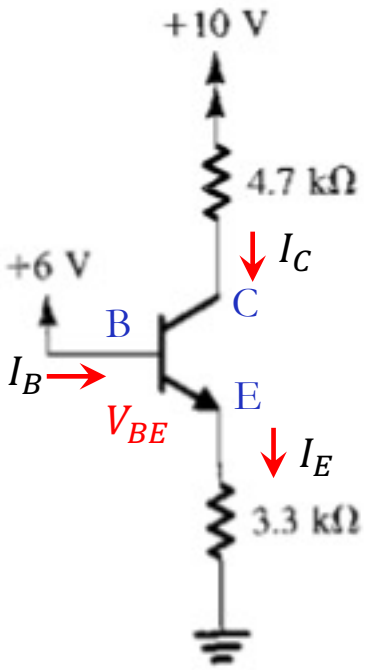
$$I_E = 3.91\ mA$$

$$V_B = 5 - (38.7\ \mu)(10\ k) = 4.6\ V < V_C$$

Active mode

Examples: Saturation Mode

11) Analyze the DC BJT circuit below. (β=100).



Circuit was checked in a previous example. It was not in active mode.

$$V_{BE} = 0.7\ V = V_B - V_E$$

$$V_B = 6\ V$$

$$V_E = 6 - 0.7 = 5.3\ V$$

$$I_E = 1.61\ mA$$

The BC junction is also forward. Then,

$$V_{BC} = 0.5\ V = V_B - V_C$$

$$V_C = 5.5\ V$$

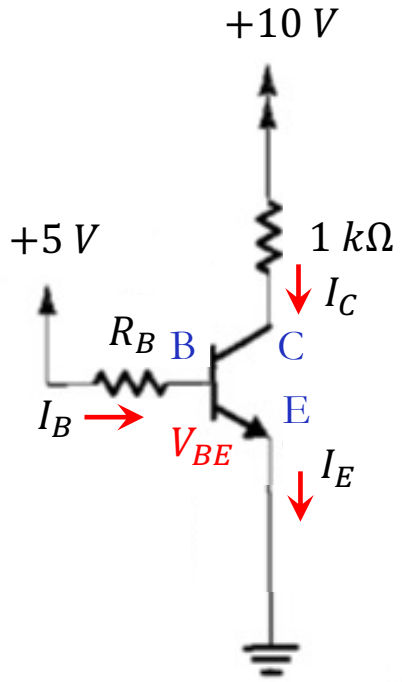
$$I_C = \frac{10 - 5.5}{4.7\ k} = 0.96\ mA$$

$$I_B = I_E - I_C = (1.61 - 0.96) = 0.65\ mA$$

$$\beta_{forced} = I_C/I_B = 1.48$$

Examples: Saturation Mode

12) Design the circuit for deep saturation with overdrive factor of 10. (β=50-100).



Saturation means both junctions are forward

$$V_E = 0\ V$$

$$V_B = 0.7\ V$$

$$V_C = 0.2\ V$$

Use the minimum β for the design.

$$I_C = \frac{10 - 0.2}{1\ k} = 9.8\ mA$$

$$I_B = \frac{I_C}{\beta_{min}} = \frac{9.8\ m}{50} = 0.196\ mA$$

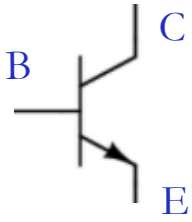
$$I_{B_sat} = 10 \times 0.196\ m = 1.96\ mA$$

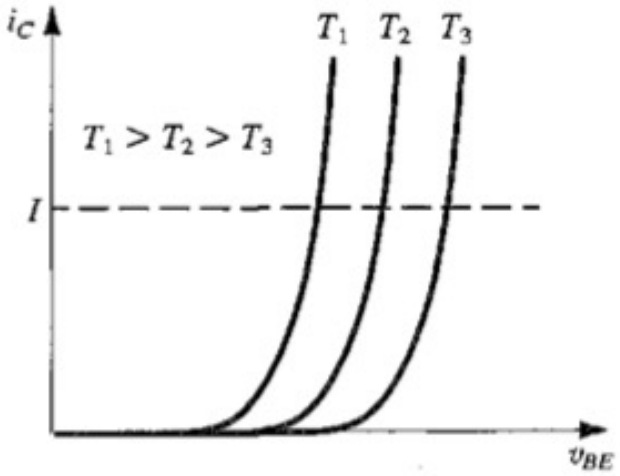
$$R_B = \frac{5 - 0.7}{1.96\ m} = 2.2\ k\Omega$$

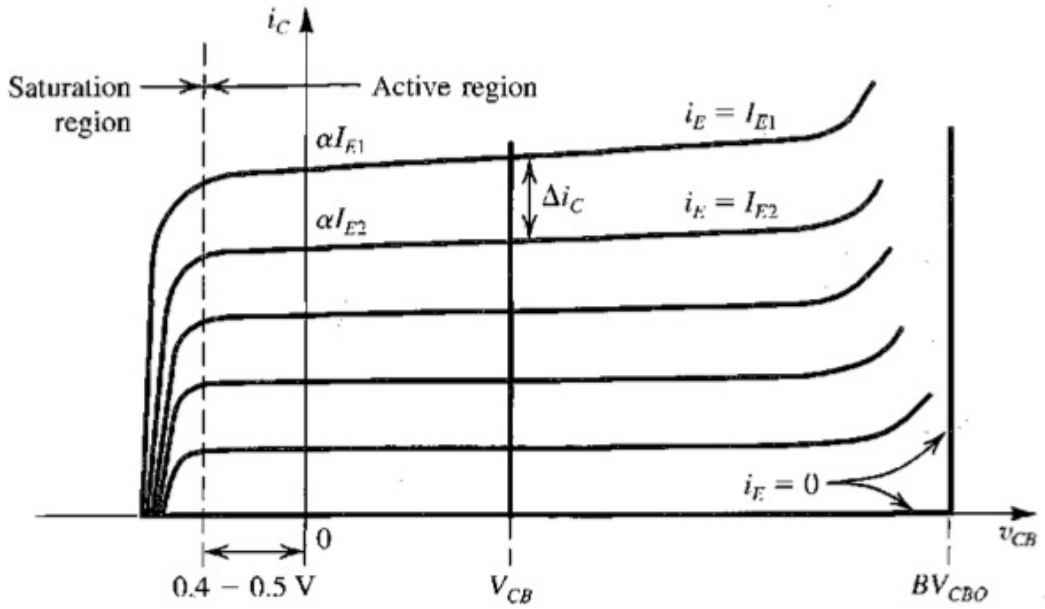
IV Characteristics

The current in the collector is a result of a forward-biased pn junction (EB junction)

$$I_C \approx I_s \, e^{V_{BE}/V_T}$$

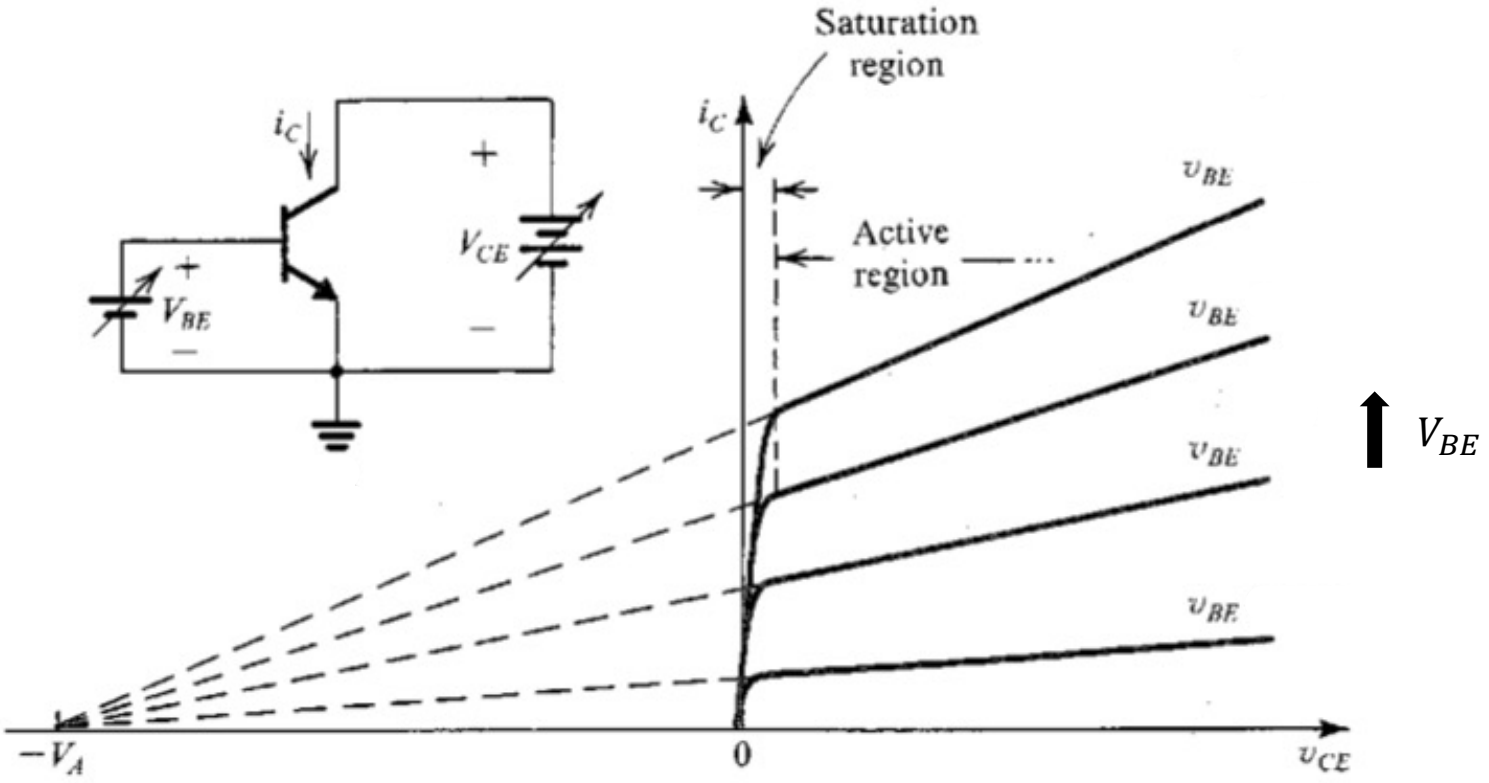






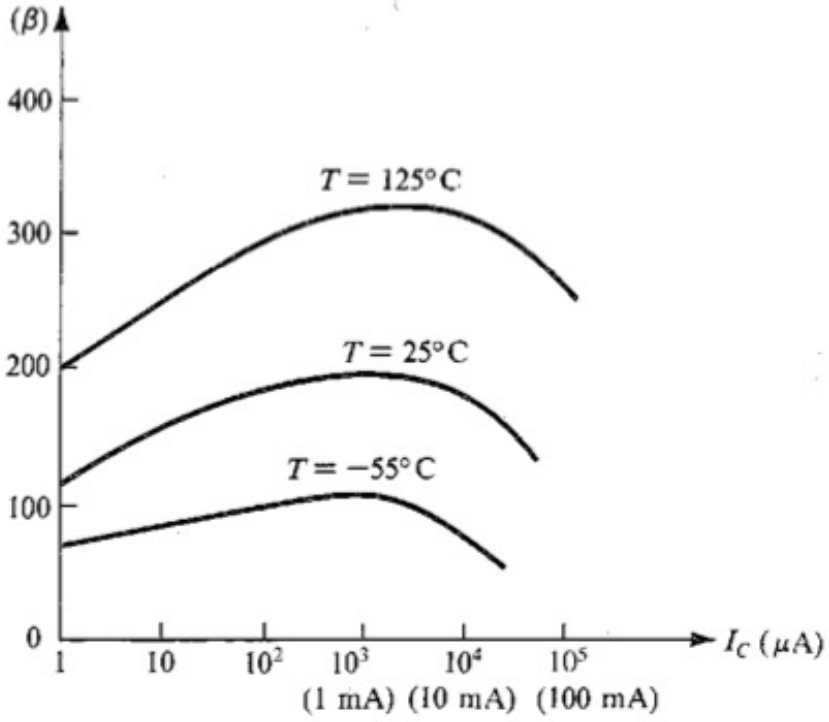
IV Characteristics

The current in the collector is a result of a forward-biased pn junction (EB junction)



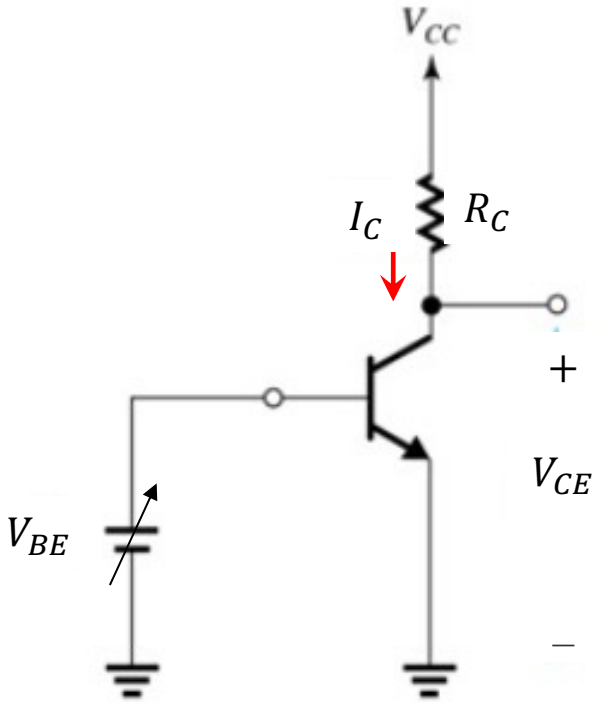
IV Characteristics

The amplification parameter of a BJT β is a function of many things; most importantly temperature.

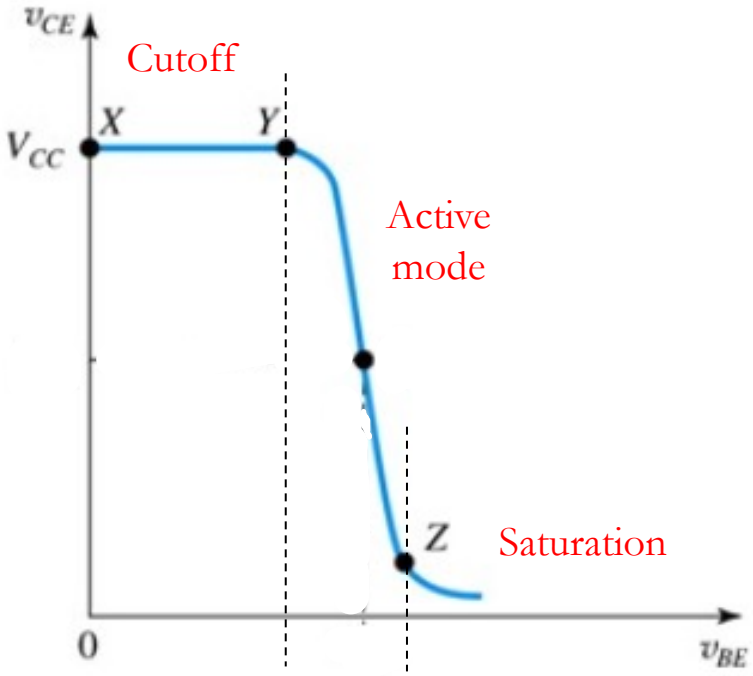


The BJT Transfer Function

The transfer function indicated the region of operation of each mode. It also help identify the possible applications.



$$V_{CE} = V_{CC} - I_C R_C$$



Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.